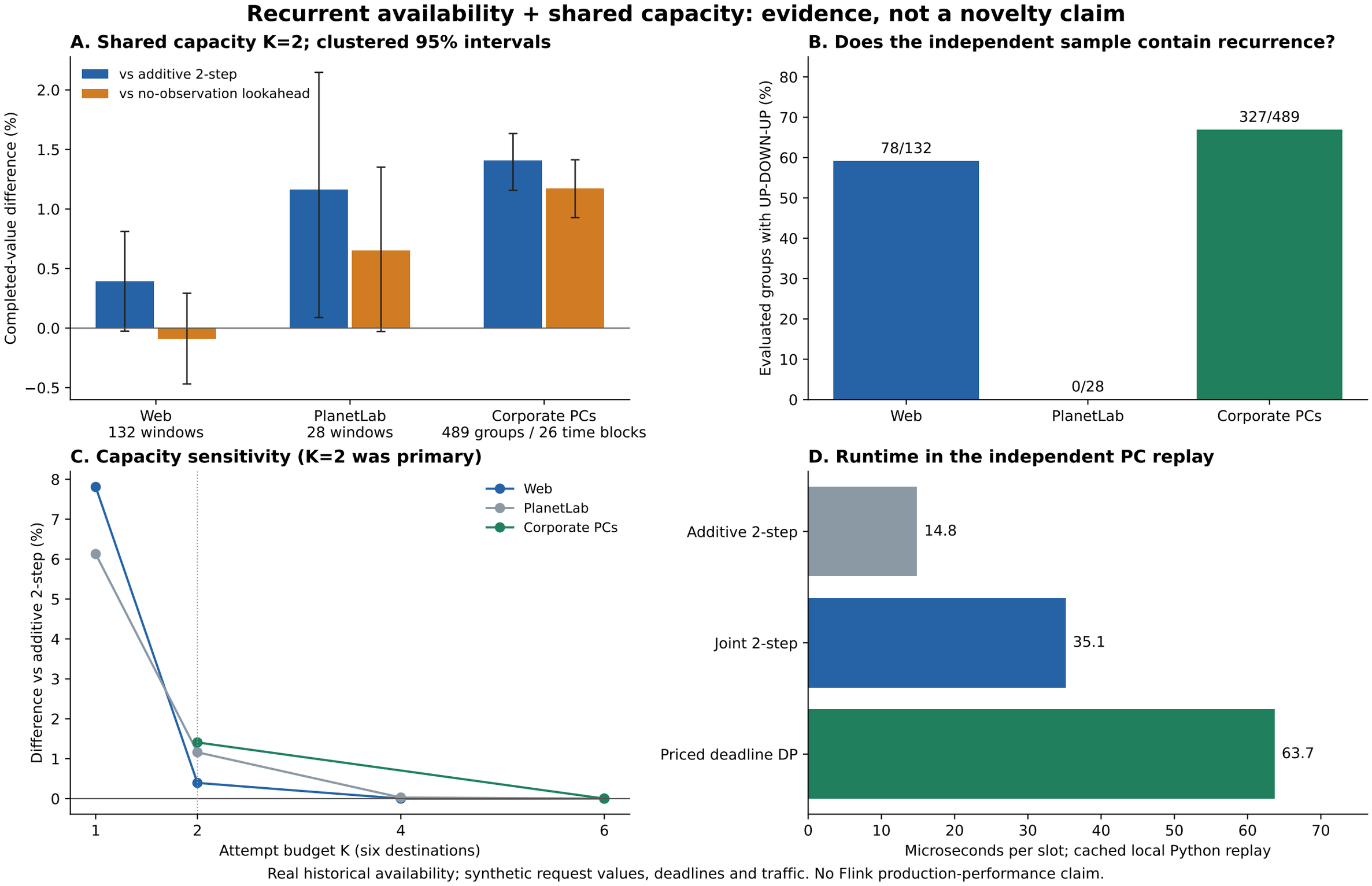
Recurrent availability + shared capacity: evidence, not a novelty claim
A. Shared capacity K=2; clustered 95% intervals
vs additive 2-step
vs no-observation lookahead
Completed-value difference (%)
2.0
1.5
1.0
0.5
0.0
−0.5
Web
132 windows
PlanetLab
28 windows
Corporate PCs
489 groups / 26 time blocks
B. Does the independent sample contain recurrence?
Evaluated groups with UP-DOWN-UP (%)
80
70
60
50
40
30
20
10
0
78/132
0/28
327/489
Web
PlanetLab
Corporate PCs
C. Capacity sensitivity (K=2 was primary)
Web
PlanetLab
Corporate PCs
Difference vs additive 2-step (%)
8
7
6
5
4
3
2
1
0
1
2
4
6
Attempt budget K (six destinations)
D. Runtime in the independent PC replay
Additive 2-step
14.8
Joint 2-step
35.1
Priced deadline DP
63.7
0
10
20
30
40
50
60
70
Microseconds per slot; cached local Python replay
Real historical availability; synthetic request values, deadlines and traffic. No Flink production-performance claim.